

DATONG, China

WMO: 534870

Lat: 40.0787N

Lon: 113.4116E

Elev: 1054

StdP: 89.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.2	-19.0	-29.1	0.3	-16.1	-27.2	0.4	-14.6	9.3	-7.7	8.4	-7.5	2.1	340	0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.0	17.7	30.5	17.2	29.1	17.1	21.5	27.0	20.7	26.0	19.9	25.1	3.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.0	16.7	23.8	19.0	15.7	23.4	18.1	14.9	22.8	68.3	27.0	65.1	26.1	62.0	25.0	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.8	-24.6	34.3	1.9	1.9	-26.0	35.7	-27.1	36.8	-28.2	37.9	-29.6	39.3		
(5)			-24.9	23.0	1.9	0.9	-26.2	23.6	-27.3	24.1	-28.4	24.6	-29.7	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-10.2	-5.6	1.8	9.9	16.1	20.8	22.8	21.0	15.8	8.4	-0.3	-8.0		
	DBStd	12.10	4.09	4.96	4.83	4.84	3.90	2.92	2.47	2.58	3.46	3.98	4.70	4.33		
	HDD10.0	2331	625	437	257	59	4	0	0	0	2	78	308	559		
	HDD18.3	4204	883	671	514	255	90	9	1	7	90	309	558	818		
	CDD10.0	1519	0	0	2	57	193	326	398	341	175	28	0	0		
	CDD18.3	349	0	0	0	2	21	85	140	89	13	0	0	0		
	CDH23.3	3333	0	0	0	54	334	883	1236	682	141	3	0	0		
	CDH26.7	1023	0	0	0	11	83	286	441	173	29	0	0	0		
(14) Wind	WSAvg	2.8	2.6	2.8	3.3	3.6	3.5	2.9	2.5	2.3	2.4	2.6	2.6	2.6		
(15) Precipitation	PrecAvg	376	2	4	9	19	29	43	102	89	49	21	7	2		
	PrecMax	579	10	18	30	66	56	77	188	232	131	48	26	6		
	PrecMin	213	0	0	0	1	2	10	22	17	8	0	0	0		
	PrecStd	87	3	4	8	15	13	19	46	48	31	14	6	2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.8	11.2	20.2	28.2	31.3	33.6	34.4	32.3	29.9	23.6	15.3	6.1		
		MCWB	-1.2	2.5	8.0	11.9	14.6	16.4	19.5	19.0	15.9	12.7	6.0	-0.3		
	2%	DB	1.7	7.6	16.1	24.5	29.0	31.4	32.3	30.1	26.7	20.6	12.4	3.2		
		MCWB	-3.2	0.8	5.5	10.5	13.5	15.8	18.6	18.9	15.4	10.6	4.5	-1.8		
	5%	DB	-0.3	5.5	13.5	22.0	27.0	29.8	30.7	28.6	24.7	18.4	9.9	1.2		
		MCWB	-4.5	-0.5	4.2	9.0	12.6	15.4	18.7	18.6	14.7	9.6	3.1	-3.2		
	10%	DB	-2.3	3.2	10.8	19.4	25.0	28.2	29.1	27.1	22.9	16.2	7.5	-0.6		
		MCWB	-5.8	-1.8	2.9	8.1	12.0	15.3	18.7	18.3	13.9	8.8	2.2	-4.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.0	3.4	8.3	13.5	16.4	19.6	22.8	22.7	19.2	13.6	7.2	0.2		
		MCDB	4.3	9.4	17.6	23.3	25.7	26.1	29.5	27.4	24.5	20.5	12.4	5.2		
	2%	WB	-2.9	1.3	6.3	11.8	15.2	18.6	21.7	21.6	17.4	12.1	5.5	-1.6		
		MCDB	1.5	7.1	14.5	21.7	24.5	25.4	27.7	26.3	22.3	17.7	10.4	2.7		
	5%	WB	-4.4	-0.2	4.9	10.3	14.1	17.9	21.0	20.7	16.1	11.0	4.0	-2.9		
		MCDB	-0.6	4.7	12.4	19.7	23.4	24.7	26.9	25.2	22.0	16.3	9.2	0.7		
	10%	WB	-5.7	-1.6	3.3	9.0	13.1	17.1	20.3	19.7	15.2	9.7	2.5	-4.1		
		MCDB	-2.5	3.0	10.2	17.3	22.1	24.1	25.9	24.4	21.1	14.8	7.4	-0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.7	13.0	13.6	13.5	12.4	10.8	10.8	11.6	11.6	11.4	11.3		
		MCDBR	14.2	15.0	17.1	17.7	17.2	15.3	13.7	13.0	14.5	14.8	14.9	13.3		
	5% WB	MCWBR	10.4	9.8	9.6	8.1	6.9	5.0	4.3	4.8	5.7	7.2	8.7	9.5		
		MCWBR	13.5	13.7	15.7	15.1	13.9	11.4	10.4	10.1	11.1	12.3	12.9	12.4		
(40) Clear-Sky Solar Irradiance	taub		0.323	0.389	0.416	0.471	0.514	0.590	0.569	0.564	0.501	0.439	0.353	0.327		
	taud		2.205	2.030	1.993	1.920	1.848	1.740	1.842	1.828	1.911	2.000	2.180	2.220		
	Ebn at Noon		819	800	825	805	778	719	730	718	734	734	771	780		
	Edh at Noon		100	139	161	185	203	227	204	201	173	142	102	92		
(44) All-Sky Solar Radiation	RadAvg		2.64	3.50	4.80	5.69	6.34	5.81	5.52	5.23	4.46	3.66	2.72	2.30		
	RadStd		0.22	0.24	0.40	0.33	0.45	0.45	0.28	0.36	0.29	0.25	0.32	0.21		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page